BANK CUSTOMER CHURN ANALYSIS

Data Analytics Portfolio Project

Project Title	Bank Customer Churn Analysis
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Project Start Date	04-10-2026
Project End Date	05-10-2026
Tools Used	Power Query, Microsoft Excel, PostgreSQL, Power BI
Dataset **Source Link**	Bank Customer Churn / Churn_Modelling.csv Kaggle – Churn Modelling

Project Objective

Analyze customer churn patterns in a bank dataset, identify customer segments associated with higher churn, and communicate practical retention-focused insights through SQL analysis and an interactive Power BI dashboard.

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1. Executive Summary

This project analyzes customer churn in a banking customer dataset containing 10,000 records. The workflow uses Power Query for preparation, Excel for exploration, PostgreSQL for deeper segment analysis, and Power BI for Dashboard/business communication.

Overall churn is 20.37%, with 2,037 customers having exited and 7,963 retained. The analysis identifies Germany, inactive membership, the 48–57 age group, and selected product/activity combinations as important areas for retention attention.

Executive Snapshot

Total Customers	10,000
Churned Customers	2,037
Retained Customers	7,963
Overall Churn Rate	20.37%
Highest clearly defined segment	Germany + Inactive - 41.08%

2. Business Problem & Background

Customer churn is an important business issue for banks because losing existing customers can reduce long-term revenue and increase the effort required to acquire replacement customers. The bank wants to understand which customer characteristics and customer segments are associated with higher churn.

Business Question

What factors are associated with customer churn, which customer segments have higher churn rates, and where should the bank focus its retention efforts?

Business Context

This is a descriptive and diagnostic analytics project rather than a predictive machinelearning model. The analysis identifies patterns and associations in the available customer data.

Expected Business Value

- Identify where churn is concentrated.
- Prioritize customer segments for retention investigation.
- Support targeted engagement strategies.

• Provide management with a clear visual summary of churn patterns.

3. Project Objectives & Analytical Questions

Objectives

• Measure overall churn.

• Compare churn across major customer characteristics.

• Identify high-churn customer segments.

• Examine the relationship between activity and churn.

• Perform deeper multi-factor analysis using SQL.

• Build a focused Power BI dashboard.

• Translate findings into practical retention recommendations.

Analytical Questions

• What is the overall churn rate?

• Which geographies have higher churn?

• Are there meaningful differences by gender and age?

• Does customer activity status relate to churn?

• Which geography/activity combinations have higher churn?

• Which age/activity segments are most exposed?

• Does number of products relate to churn, particularly for inactive customers?

• Which segments deserve retention attention?

4. Dataset Overview & Data Dictionary

The project uses the standard Kaggle Bank Customer Churn dataset, commonly provided as Churn_Modelling.csv. The working dataset contains 10,000 customer records and 13 columns. Exited is the target variable: 0 = retained and 1 = churned.

Source: Kaggle

Dataset File: Churn_Modelling.csv

Source Link: Kaggle – Churn Modelling Dataset

Column	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score
Geography	Customer country/region
Gender	Customer gender
Age	Customer age
Tenure	Length of relationship
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit card indicator
IsActiveMember	Active membership indicator
EstimatedSalary	Estimated salary
Exited	Churn indicator: 0 retained, 1 churned

Target Variable

Exited = 1 indicates churned customers; Exited = 0 indicates retained customers.

5. Tools & Methodology

Tools Used

- Microsoft Excel / Power Query - data preparation and exploratory analysis
- PostgreSQL - deeper business and segment analysis
- Power BI - interactive dashboard and communication
- GitHub - portfolio/project presentation

End-to-End Workflow

Power Query Cleaning	Excel Exploration	SQL Deep Analysis	Power BI Dashboard

Analytical Approach

The workflow intentionally separates responsibilities between tools. Cleaning is performed in Power Query, exploratory pattern discovery in Excel, deeper multifactor analysis in SQL, and final communication in Power BI. This avoids unnecessarily repeating the same analysis in every tool.

6. Data Preparation & Cleaning

Power Query was used to prepare the dataset before analysis. The goal was to ensure that the data was structurally consistent without introducing unnecessary transformations.

- Imported the CSV through Excel Power Query.
- Checked column names and structure.
- Verified data types.
- Checked missing values.
- Checked duplicate records.
- Checked categorical values for inconsistencies.
- Checked numerical fields for invalid values.

• Applied only necessary cleaning transformations.

• Loaded the cleaned dataset for analysis.

Data Quality Principle

No analytical conclusion was based on an assumed data-quality issue. Cleaning focused on validation and consistency. Identifier fields such as CustomerId and Surname were retained but were not treated as analytical drivers.

Cleaning Status

Final working dataset: 10,000 customer records and 13 columns.

7. Exploratory Data Analysis

Excel was used for initial exploration and pattern discovery. Only important businessrelevant findings were retained for the final project.

Overall Churn

10,000 customers were analyzed. 2,037 customers churned and 7,963 were retained, giving an overall churn rate of 20.37%.

Geography

Germany had the highest churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain.

Gender

Female customers had a 25.07% churn rate compared with 16.46% for male customers.

Age

The 48–57 age group showed the highest substantial age-group churn rate at 55.26%. Extremely small age groups were not treated as headline findings.

Activity

Inactive members had a 26.85% churn rate versus 14.27% for active members.

Number of Products

One product customers had a 27.71% churn rate compared with 7.58% for twoproduct customers. Very small 3 and 4 product groups require cautious interpretation. **Tenure**

Tenure showed relatively smaller differences across most tenure values, so it was not treated as a primary headline finding.

8. SQL Business Analysis

PostgreSQL was used to investigate combinations of variables that provide stronger business segmentation than single-variable exploration.

8.1 Geography + Active Membership

Germany + inactive customers had the highest clearly defined segment churn at 41.08%. Germany + active was 23.72%. Spain + inactive was 23.35%, France + inactive was 21.13%, France + active was 11.50%, and Spain + active was 10.75%.

Interpretation

Inactive customers had higher churn in every geography. Germany's inactive segment is therefore an important retention priority.

8.2 Age Group + Active Membership

Customers aged 48–57 who were inactive had 73.27% churn, compared with 38.73% among active customers in the same age group.

Interpretation

The activity difference is particularly strong within this age segment. Other extreme results from very small groups should be treated as investigation flags.

8.3 Number of Products + Active Membership

One-product + inactive customers had approximately 36% churn versus approximately 18% for one-product + active customers. Two-product churn was much lower. The 4product group had 100% churn but only 60 customers, so it should not be generalized.

9. Power BI Dashboard

Power BI was used to communicate the validated analysis through two focused dashboard pages.

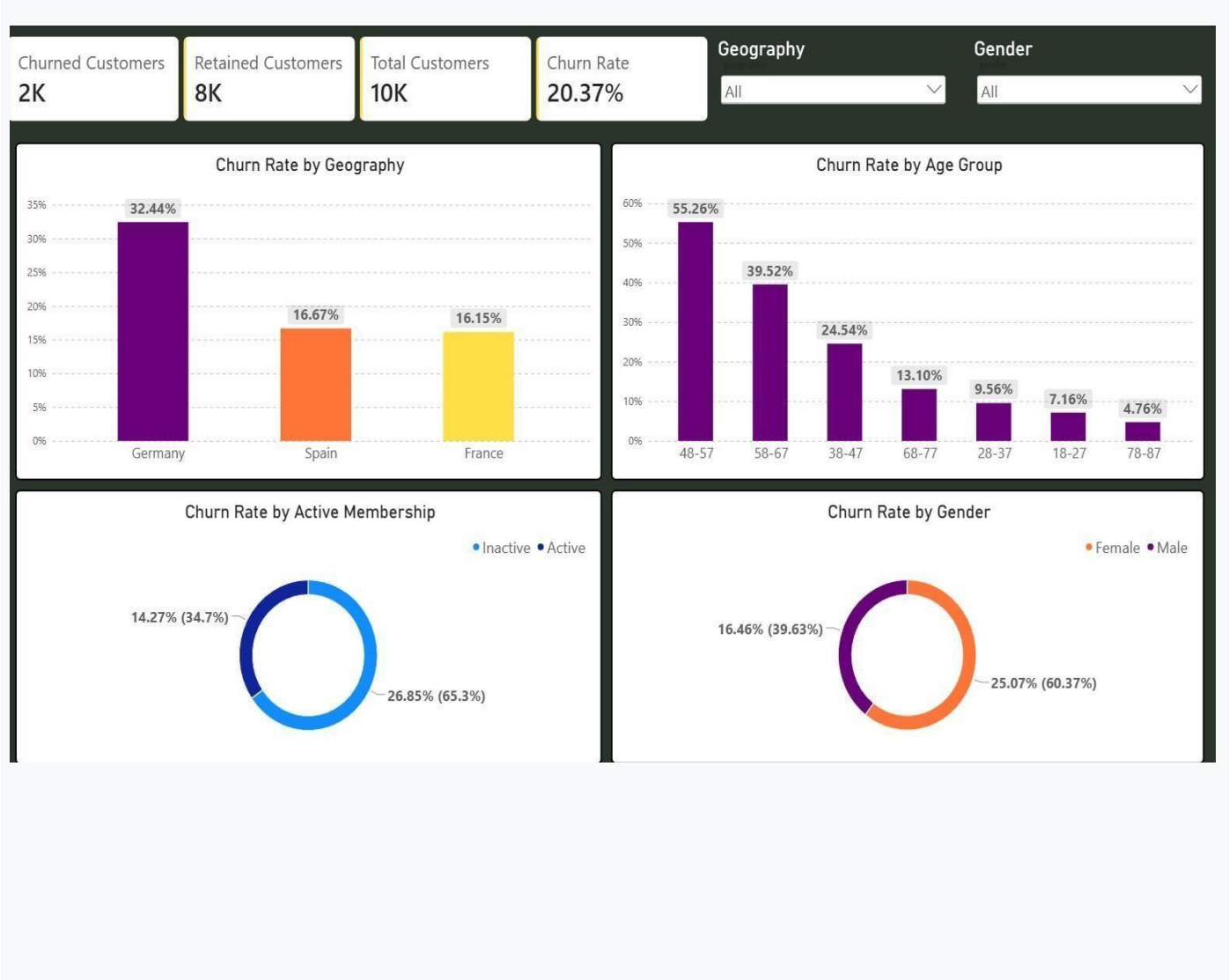
Page 1 - Bank Customer Churn Overview

KPI cards show Total Customers (10,000), Churned Customers (2,037), Retained Customers (7,963), and Churn Rate (20.37%). Key visuals cover geography, gender, age and active membership. Geography and gender slicers provide interactive filtering.

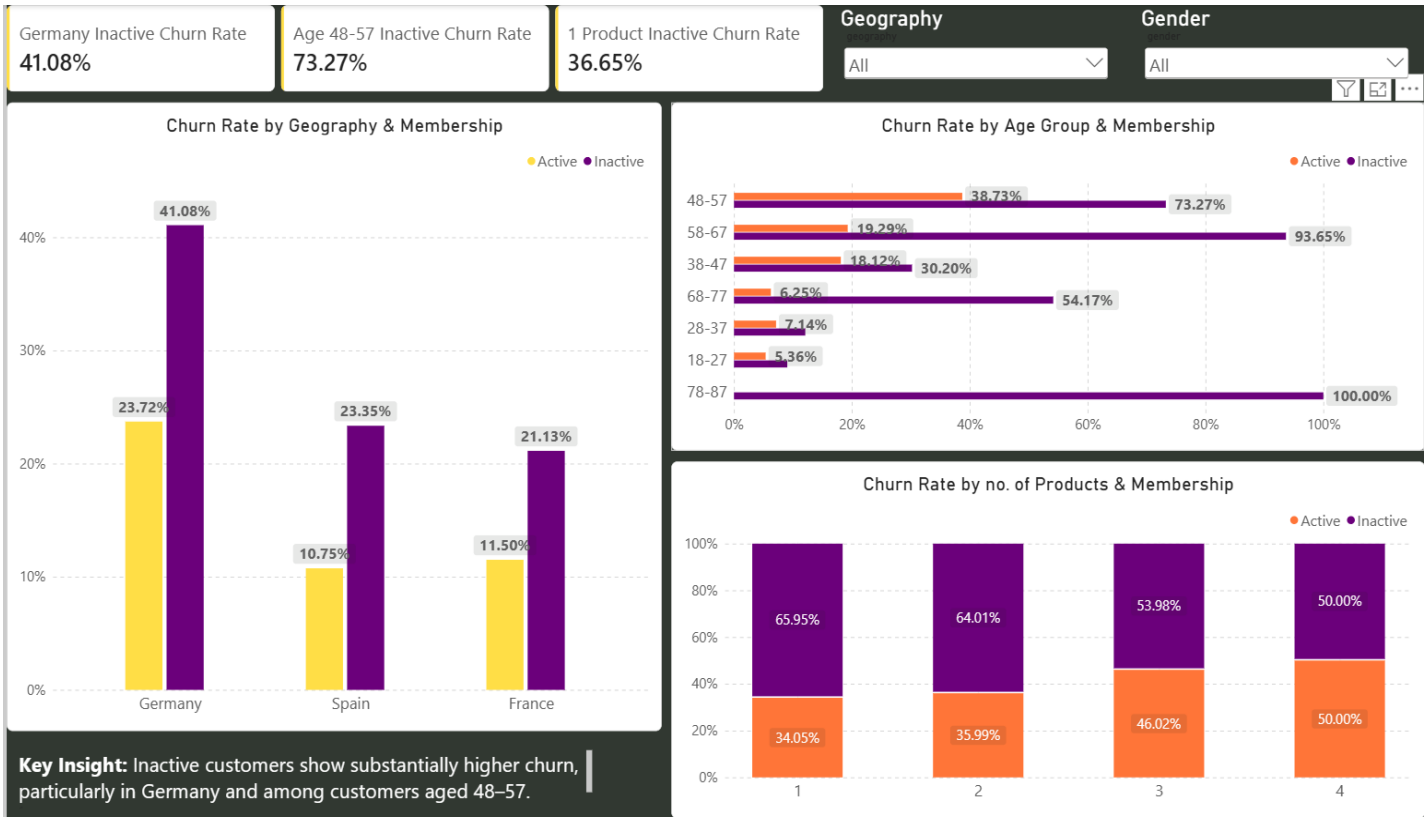
Page 2 - High-Risk Customer Segments

The second page focuses on Germany + inactive (41.08%), age 48–57 + inactive (73.27%), and one-product + inactive (36%). Comparative visuals show how activity changes churn across these segments.

Dashboard Graph 1



Dashboard Graph 2



10. Key Findings & Business Recommendations

Key Findings

- Overall churn is 20.37%.
- Germany has the highest overall churn at 32.44%.
- Inactive customers have materially higher churn than active customers.
- Germany + inactive is the highest clearly defined segment at 41.08%.
- Age 48–57 + inactive reaches 73.27% churn.
- One-product customers have higher churn than two-product customers, with inactivity increasing the difference.
- Very high rates in small 3 and 4 product groups should be investigated rather than generalized. The bank should **look more closely at those 60 customers first** and check whether there is a genuine pattern or something unusual about that group.

Business Recommendations

- Prioritize re-engagement of inactive customers.
- Investigate why Germany has higher churn and whether service, pricing, competition or product fit differs.
- Create targeted retention journeys for inactive customers aged 48–57.
- Investigate one-product customers and whether stronger engagement is associated with retention.
- Investigate small high-risk customer groups before making major business decisions.
- Track whether retention campaigns successfully reactivate customers and reduce future churn.

11. Limitations & Conclusion

Limitations

- The dataset is a snapshot rather than a customer-level time series.
- It does not provide detailed transaction history or activity timelines.
- It does not establish why a customer churned.
- Associations should not be interpreted as causation.
- Some extreme churn rates occur in small segments.
- Important real-world drivers such as complaints, satisfaction, service interactions and competitor activity are not included.

Conclusion

Customer churn is unevenly distributed across the customer base. Germany, inactive customers and selected age/product segments show substantially higher churn. The

strongest practical pattern is the association between inactivity and churn, particularly in Germany and among customers aged 48–57.

The analysis provides a focused starting point for retention strategy and further investigation. It demonstrates a complete Data Analytics workflow from cleaning and exploration to SQL segmentation and Power BI communication.

Final Business Takeaway

Retention efforts should begin with the highest-risk, sufficiently sized segments especially inactive customers in Germany and inactive customers aged 48–57 while small extreme segments should be investigated before broad action.

12. Project Deliverables / Appendix

Project Deliverables

- Cleaned dataset
- Excel exploratory analysis
- PostgreSQL - SQL queries and results
- Power BI dashboard (.pbix)
- Final project report
- Dashboard screenshots
- GitHub README

Important SQL Analysis

Core segmentation logic: group customers by Geography and IsActiveMember, count total customers and churned customers, and calculate churn rate as churned customers divided by total customers.